

OpenStax Calculus Volume 3

Section 2.5 Exercises: Equations of Lines and Planes in Space

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In the following exercises, points P and Q are given. Let L be the line passing through points P and Q .

- (a) Find the vector equation of line L .
- (b) Find parametric equations of line L .
- (c) Find symmetric equations of line L .
- (d) Find parametric equations of the line segment determined by P and Q .

- 1. $P(-3, 5, 9)$, $Q(4, -7, 2)$
- 2. $P(4, 0, 5)$, $Q(2, 3, 1)$
- 3. $P(-1, 0, 5)$, $Q(4, 0, 3)$
- 4. $P(7, -2, 6)$, $Q(-3, 0, 6)$

For the following exercises, point P and vector $\mathbf{v}$ are given. Let L be the line passing through point P with direction $\mathbf{v}$.

- (a) Find parametric equations of line L .
- (b) Find symmetric equations of line L .
- (c) Find the intersection of the line with the xy -plane.

- 5. $P(1, -2, 3)$, $\mathbf{v} = \langle 1, 2, 3 \rangle$
- 6. $P(3, 1, 5)$, $\mathbf{v} = \langle 1, 1, 1 \rangle$
- 7. $P(3, 1, 5)$, $\mathbf{v} = \overrightarrow{QR}$, where $Q(2, 2, 3)$ and $R(3, 2, 3)$
- 8. $P(2, 3, 0)$, $\mathbf{v} = \overrightarrow{QR}$, where $Q(0, 4, 5)$ and $R(0, 4, 6)$

For the following exercises, line L is given.

- (a) Find point P that belongs to the line and direction vector $\mathbf{v}$ of the line. Express $\mathbf{v}$ in component form.
- (b) Find the distance from the origin to line L .

9. $x = 1 + t, y = 3 + t, z = 5 + 4t, t \in \mathbb{R}$
10. $-x = y + 1, z = 2$
11. Find the distance between point $A(-3, 1, 1)$ and the line of symmetric equations $x = -y = -z$.
12. Find the distance between point $A(4, 2, 5)$ and the line of parametric equations $x = -1 - t, y = -t, z = 2, t \in \mathbb{R}$.

For the following exercises, lines L_1 and L_2 are given.

- (a) Verify whether lines L_1 and L_2 are parallel.
- (b) If the lines L_1 and L_2 are parallel, then find the distance between them.

13. $L_1 : x = 1 + t, y = t, z = 2 + t, t \in \mathbb{R}, L_2 : x - 3 = y - 1 = z - 3$
14. $L_1 : x = 2, y = 1, z = t, L_2 : x = 1, y = 1, z = 2 - 3t, t \in \mathbb{R}$
15. Show that the line passing through points $P(3, 1, 0)$ and $Q(1, 4, -3)$ is perpendicular to the line with equations $x = 3 + 3t, y = 1 + 8t, z = 6t, t \in \mathbb{R}$.
16. Are the lines of equations $x = -2 + 2t, y = -6, z = 2 + 6t$ and $x = -1 + t, y = 1 + t, z = t, t \in \mathbb{R}$, perpendicular to each other?
17. Find the point of intersection of the lines of equations $x = -2y = 3z$ and $x = -5 - t, y = -1 + t, z = t - 11, t \in \mathbb{R}$.
18. Find the intersection point of the x -axis with the line of parametric equations $x = 10 + t, y = 2 - 2t, z = -3 + 3t, t \in \mathbb{R}$.

For the following exercises, lines L_1 and L_2 are given. Determine whether the lines are equal, parallel but not equal, skew, or intersecting.

19. $L_1 : x = y - 1 = -z$ and $L_2 : x - 2 = -y = \frac{z}{2}$
20. $L_1 : x = 2t, y = 0, z = 3, t \in \mathbb{R}$ and $L_2 : x = 0, y = 8 + s, z = 7 + s, s \in \mathbb{R}$
21. $L_1 : x = -1 + 2t, y = 1 + 3t, z = 7t, t \in \mathbb{R}$ and $L_2 : x - 1 = \frac{2}{3}(y - 4) = \frac{2}{7}z - 2$
22. $L_1 : 3x = y + 1 = 2z$ and $L_2 : x = 6 + 2t, y = 17 + 6t, z = 9 + 3t, t \in \mathbb{R}$
23. Consider line L of symmetric equations $x - 2 = -y = \frac{z}{2}$ and point $A(1, 1, 1)$.
 - (a) Find parametric equations for a line parallel to L that passes through point A .
 - (b) Find symmetric equations of a line skew to L and that passes through point A .
 - (c) Find symmetric equations of a line that intersects L and passes through point A .
24. Consider line L of parametric equations $x = t, y = 2t, z = 3, t \in \mathbb{R}$.
 - (a) Find parametric equations for a line parallel to L that passes through the origin.
 - (b) Find parametric equations of a line skew to L that passes through the origin.
 - (c) Find symmetric equations of a line that intersects L and passes through the origin.

For the following exercises, point P and vector $\mathbf{n}$ are given.

- (a) Find the scalar equation of the plane that passes through P and has normal vector $\mathbf{n}$.

(b) Find the general form of the equation of the plane that passes through P and has normal vector $\mathbf{n}$.

25. $P(0, 0, 0)$, $\mathbf{n} = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$

26. $P(3, 2, 2)$, $\mathbf{n} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$

27. $P(1, 2, 3)$, $\mathbf{n} = \langle 1, 2, 3 \rangle$

28. $P(0, 0, 0)$, $\mathbf{n} = \langle -3, 2, -1 \rangle$

For the following exercises, the equation of a plane is given.

(a) Find normal vector $\mathbf{n}$ to the plane. Express $\mathbf{n}$ using standard unit vectors.

(b) Find the intersections of the plane with the coordinate axes.

(c) Sketch the plane.

29. [T] $4x + 5y + 10z - 20 = 0$

30. $3x + 4y - 12 = 0$

31. $3x - 2y + 4z = 0$

32. $x + z = 0$

33. Given point $P(1, 2, 3)$ and vector $\mathbf{n} = \mathbf{i} + \mathbf{j}$, find point Q on the x -axis such that $\vec{PQ}$ and $\mathbf{n}$ are orthogonal.

34. Show there is no plane perpendicular to $\mathbf{n} = \mathbf{i} + \mathbf{j}$ that passes through points $P(1, 2, 3)$ and $Q(2, 3, 4)$.

35. Find parametric equations of the line passing through point $P(-2, 1, 3)$ that is perpendicular to the plane of equation $2x - 3y + z = 7$.

36. Find symmetric equations of the line passing through point $P(2, 5, 4)$ that is perpendicular to the plane of equation $2x + 3y - 5z = 0$.

37. Show that line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{4}$ is parallel to plane $x - 2y + z = 6$.

38. Find the real number α such that the line of parametric equations $x = t, y = 2 - t, z = 3 + t$, $t \in \mathbb{R}$ is parallel to the plane of equation $\alpha x + 5y + z - 10 = 0$.

For the following exercises, points P , Q , and R are given.

(a) Find the general equation of the plane passing through P , Q , and R .

(b) Write the vector equation $\mathbf{n} \cdot \vec{PS} = 0$ of the plane at a., where $S(x, y, z)$ is an arbitrary point of the plane.

(c) Find parametric equations of the line passing through the origin that is perpendicular to the plane passing through P , Q , and R .

39. $P(1, 1, 1)$, $Q(2, 4, 3)$, and $R(-1, -2, -1)$

40. $P(-2, 1, 4)$, $Q(3, 1, 3)$, and $R(-2, 1, 0)$

41. Consider the planes of equations $x + y + z = 1$ and $x + z = 0$.

- (a) Show that the planes intersect.
 - (b) Find symmetric equations of the line passing through point $P(1, 4, 6)$ that is parallel to the line of intersection of the planes.
42. Consider the planes of equations $-y + z - 2 = 0$ and $x - y = 0$.
- (a) Show that the planes intersect.
 - (b) Find parametric equations of the line passing through point $P(-8, 0, 2)$ that is parallel to the line of intersection of the planes.
43. Find the scalar equation of the plane that passes through point $P(-1, 2, 1)$ and is perpendicular to the line of intersection of planes $x + y - z - 2 = 0$ and $2x - y + 3z - 1 = 0$.
44. Find the general equation of the plane that passes through the origin and is perpendicular to the line of intersection of planes $-x + y + 2 = 0$ and $z - 3 = 0$.
45. Determine whether the line of parametric equations $x = 1 + 2t, y = -2t, z = 2 + t, t \in \mathbb{R}$ intersects the plane with equation $3x + 4y + 6z - 7 = 0$. If it does intersect, find the point of intersection.
46. Determine whether the line of parametric equations $x = 5, y = 4 - t, z = 2t, t \in \mathbb{R}$ intersects the plane with equation $2x - y + z = 5$. If it does intersect, find the point of intersection.
47. Find the distance from point $P(1, 5, -4)$ to the plane of equation $3x - y + 2z - 6 = 0$.
48. Find the distance from point $P(1, -2, 3)$ to the plane of equation $(x - 3) + 2(y + 1) - 4z = 0$.

For the following exercises, the equations of two planes are given.

- (a) Determine whether the planes are parallel, orthogonal, or neither.
 - (b) If the planes are neither parallel nor orthogonal, then find the measure of the angle between the planes. Express the answer in degrees rounded to the nearest integer.
49. [T] $x + y + z = 0, 2x - y + z - 7 = 0$
50. $5x - 3y + z = 4, x + 4y + 7z = 1$
51. $x - 5y - z = 1, 5x - 25y - 5z = -3$
52. [T] $x - 3y + 6z = 4, 5x + y - z = 4$
53. Show that the lines of equations $x = t, y = 1 + t, z = 2 + t, t \in \mathbb{R}$, and $\frac{x}{2} = \frac{y-1}{3} = z - 3$ are skew, and find the distance between them.
54. Show that the lines of equations $x = -1 + t, y = -2 + t, z = 3t, t \in \mathbb{R}$, and $x = 5 + s, y = -8 + 2s, z = 7s, s \in \mathbb{R}$ are skew, and find the distance between them.
55. Consider point $C(-3, 2, 4)$ and the plane of equation $2x + 4y - 3z = 8$.
- (a) Find the radius of the sphere with center C tangent to the given plane.
 - (b) Find point P of tangency.
56. Consider the plane of equation $x - y - z - 8 = 0$.
- (a) Find an equation of the sphere with center C at the origin that is tangent to the given plane.

- (b) Find parametric equations of the line passing through the origin and the point of tangency.
57. Two children are playing with a ball. The girl throws the ball to the boy. The ball travels in the air, curves 3 ft to the right, and falls 5 ft away from the girl (see the following figure). If the plane that contains the trajectory of the ball is perpendicular to the ground, find its equation.
[Figure: This figure is the image of two children throwing a ball. The path of the ball is represented with an arc. The distance from the child throwing the ball to the point where the ball hits is 5 feet. The distance from the second child to where the ball hits is 3 feet.]
58. [T] John allocates d dollars to consume monthly three goods of prices a , b , and c . In this context, the budget equation is defined as $ax + by + cz = d$, where $x \geq 0$, $y \geq 0$, and $z \geq 0$ represent the number of items bought from each of the goods. The budget set is given by $\{(x, y, z) | ax + by + cz \leq d, x \geq 0, y \geq 0, z \geq 0\}$, and the budget plane is the part of the plane of equation $ax + by + cz = d$ for which $x \geq 0$, $y \geq 0$, and $z \geq 0$. Consider $a = \$8$, $b = \$5$, $c = \$10$, and $d = \$500$.
- (a) Use a CAS to graph the budget set and budget plane.
- (b) For $z = 25$, find the new budget equation and graph the budget set in the same system of coordinates.
59. [T] Consider $\mathbf{r}(t) = \langle \sin t, \cos t, 2t \rangle$ the position vector of a particle at time $t \in [0, 3]$, where the components of $\mathbf{r}$ are expressed in centimeters and time is measured in seconds. Let $\vec{OP}$ be the position vector of the particle after 1 sec.
- (a) Determine the velocity vector $\mathbf{v}(1)$ of the particle after 1 sec.
- (b) Find the scalar equation of the plane that is perpendicular to $\mathbf{v}(1)$ and passes through point P . This plane is called the normal plane to the path of the particle at point P .
- (c) Use a CAS to visualize the path of the particle along with the velocity vector and normal plane at point P .
60. [T] A solar panel is mounted on the roof of a house. The panel may be regarded as positioned at the points of coordinates (in meters) $A(8, 0, 0)$, $B(8, 18, 0)$, $C(0, 18, 8)$, and $D(0, 0, 8)$ (see the following figure).
[Figure: This figure is a picture of a rectangular solar grid on a roof. The corners of the rectangle are labeled A, B, C, D. There are two vectors, the first is from A to D. The second is from A to B.]
- (a) Find the general form of the equation of the plane that contains the solar panel by using points A , B , and C , and show that its normal vector is equivalent to $\vec{AB} \times \vec{AD}$.
- (b) Find parametric equations of line L_1 that passes through the center of the solar panel and has direction vector $\mathbf{s} = \frac{1}{\sqrt{3}}\mathbf{i} + \frac{1}{\sqrt{3}}\mathbf{j} + \frac{1}{\sqrt{3}}\mathbf{k}$, which points toward the position of the Sun at a particular time of day.
- (c) Find symmetric equations of line L_2 that passes through the center of the solar panel and is perpendicular to it.
- (d) Determine the angle of elevation of the Sun above the solar panel by using the angle between lines L_1 and L_2 .